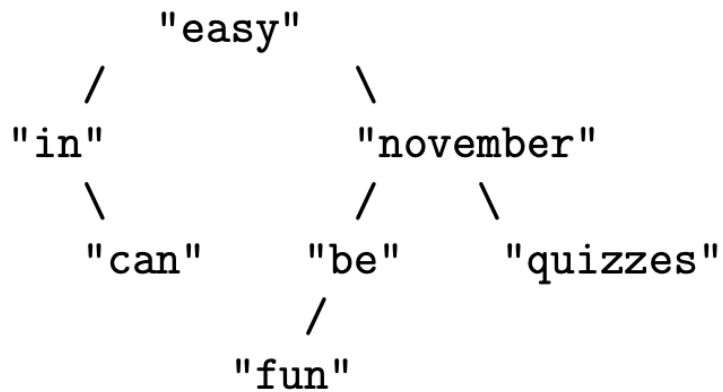


## Tutorial Sheet: Tree

1. Consider the following binary tree:



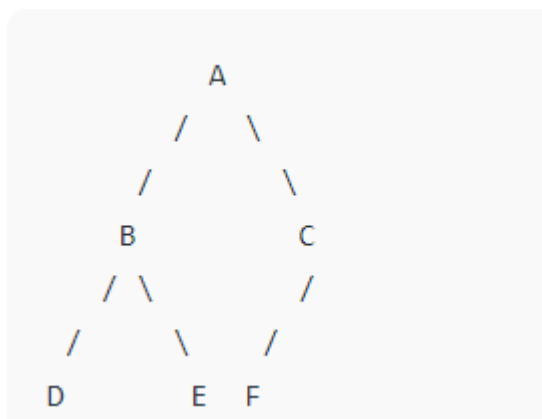
- (a) What is the order of nodes visited in a pre-order traversal?  
(b) What is the order of nodes visited in a post-order traversal?

Answer:

- (a) easy, in, can, november, be, fun, quizzes  
(b) can, in, fun, be, quizzes, november, easy

2. a. Draw the binary tree whose in-order traversal is DBEAFC and whose pre-order traversal is ABDECF.

Answer:

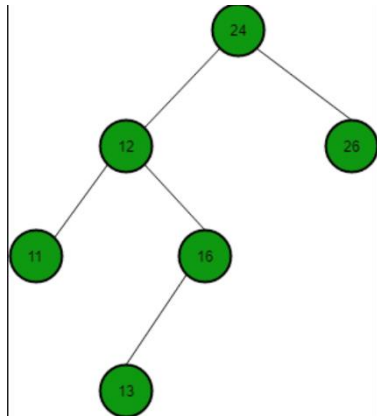


- b. Write down its post order traversal

Answer:

DEBFCA

3. Consider the following binary search tree:



If we remove the root node, which of the node from the left subtree will be the new root?

- (A) 11
- (B) 12
- (C) 13
- (D) 16

**Ans: D**

4. Which of the following traversals is sufficient to construct BST from given traversals

1) Inorder 2) Preorder 3) Postorder

- A. Any one of the given three traversals is sufficient,
- B. Either 2 or 3 is sufficient
- C. 2 and 3
- D. 1 and 3

**Ans: B**

When we know either preorder or postorder traversal, we can construct the BST. Note that we can always sort the given traversal and get the inorder traversal. Inorder traversal of BST is always sorted.

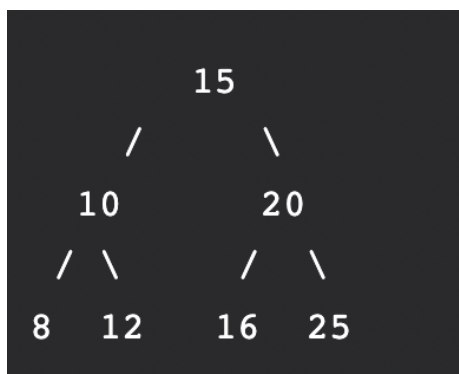
5. Consider the following code snippet in C. The function print() receives root of a Binary Search Tree (BST) and a positive integer k as arguments.

```
// A BST node
struct node {
    int data;
    struct node *left, *right;
};

int count = 0;

void print(struct node *root, int k)
{
    if (root != NULL && count <= k)
    {
        print(root->right, k);
        count++;
        if (count == k)
            printf("%d ", root->data);
        print(root->left, k);
    }
}
```

What is the output of print(root, 3) where root represent root of the following BST?



- (A) 10
- (B) 16
- (C) 20
- (D) 20 10

Answer: (B)

Explanation: The code mainly finds out k'th largest element in BST, see [K'th Largest Element in BST](#) for details.

6. What are the worst-case complexities of insertion and deletion of a key in a binary search tree?

- A,  $\Theta(\log n)$  for both insertion and deletion
- B,  $\Theta(n)$  for both insertion and deletion
- C,  $\Theta(n)$  for insertion and  $\Theta(\log n)$  for deletion
- D,  $\Theta(\log n)$  for insertion and  $\Theta(n)$  for deletion

Ans: B

7. While inserting the elements 71, 65, 84, 69, 67, 83 in an empty binary search tree (BST) in the sequence shown, the element in the lowest level is

- A, 65
- B, 67
- C, 69
- D, 83

Ans: B

Solution: <https://www.geeksforgeeks.org/gate-gate-cs-2015-set-3-question-23/>

8. What is common in three different types of traversals (Inorder, Preorder and Postorder)?

- a) Root is visited before right subtree
- b) Left subtree is always visited before right subtree
- c) Root is visited after left subtree
- d) All of the above
- e) None of the above

Ans: b

**Explanation:**

The order of inorder traversal is LEFT ROOT RIGHT The order of preorder traversal is ROOT LEFT RIGHT The order of postorder traversal is LEFT RIGHT ROOT In all three traversals, LEFT is traversed before RIGHT

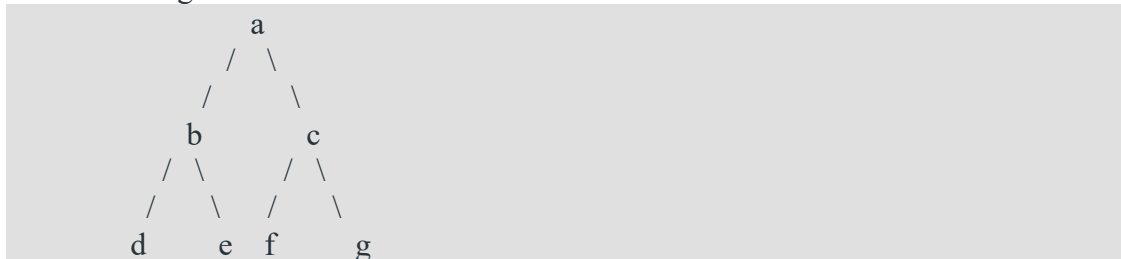
9. The inorder and preorder traversal of a binary tree are d b e a f c g and a b d e c f g, respectively. The postorder traversal of the binary tree is:
- a) d e b f g c a
  - b) e d b g f c a
  - c) e d b f g c a
  - d) d e f g b c a

**Ans: a**

**Explanation:**

Please see this link for more details <http://www.geeksforgeeks.org/construct-tree-from-given-inorder-and-preorder-traversal/>

Below is the given tree.



10. Which traversal of tree resembles the breadth first search of the graph?
- a) Preorder
  - b) Inorder
  - c) Postorder
  - d) Level order

**Ans: d**

**Explanation:**

Breadth first search visits all the neighbors first and then deepens into each neighbor one by one. The level order traversal of the tree also visits nodes on the current level and then goes to the next level.

11. Which of the following tree traversal uses a queue data structure?
- a) Preorder
  - b) Inorder
  - c) Postorder
  - d) Level order

**Ans: d**

**Explanation:**

Level order traversal uses a queue data structure to visit the nodes level by level.

12. A binary search tree contains the numbers 1, 2, 3, 4, 5, 6, 7, 8. When the tree is traversed in pre-order and the values in each node printed out, the sequence of values obtained is 5, 3, 1, 2, 4, 6, 8, 7. If the tree is traversed in post-order, the sequence obtained would be

a) 8, 7, 6, 5, 4, 3, 2, 1

b) 1, 2, 3, 4, 8, 7, 6, 5

c) 2, 1, 4, 3, 6, 7, 8, 5

d) 2, 1, 4, 3, 7, 8, 6, 5

**Ans: d**

**Explanation:**

Please see this link for more details <http://www.geeksforgeeks.org/construct-tree-from-given-inorder-and-preorder-traversal/>